Digital Systems Laboratory - 6

Designing a Traffic Signal Counter

You are required to use the SEVEN segment display that you designed in experiment-4 (one 7447 and a multiplexer) and convert it to counter at a traffic signal. It should start at 15 and count down to 0 at a rate which is as close to 1 Hz as possible. Remember you cannot house a signal generator in a traffic light set up and neither is it economically viable to use such an expensive instrument for such a simple operation. Is it possible to make this count down at 1 Hz for a smaller range of values? In addition you need to configure the traffic signal MSB to display any arbitray n and $n+1$.

You have the following following components available for your use-

1. $555 - timer \times 1$
2. $74LS163 \times 2$
3. Resistors available in the lab
4. Capacitors available in the lab.

You have to read the data sheet of the ICs and figure out the connections to implement your design.

Generating a Pulse

You can use this information to generate a pulse using a 555 timer.

Frequency of oscillations is

$$f = \frac{1}{\ln(2)(R_1 + 2R_2)C}$$

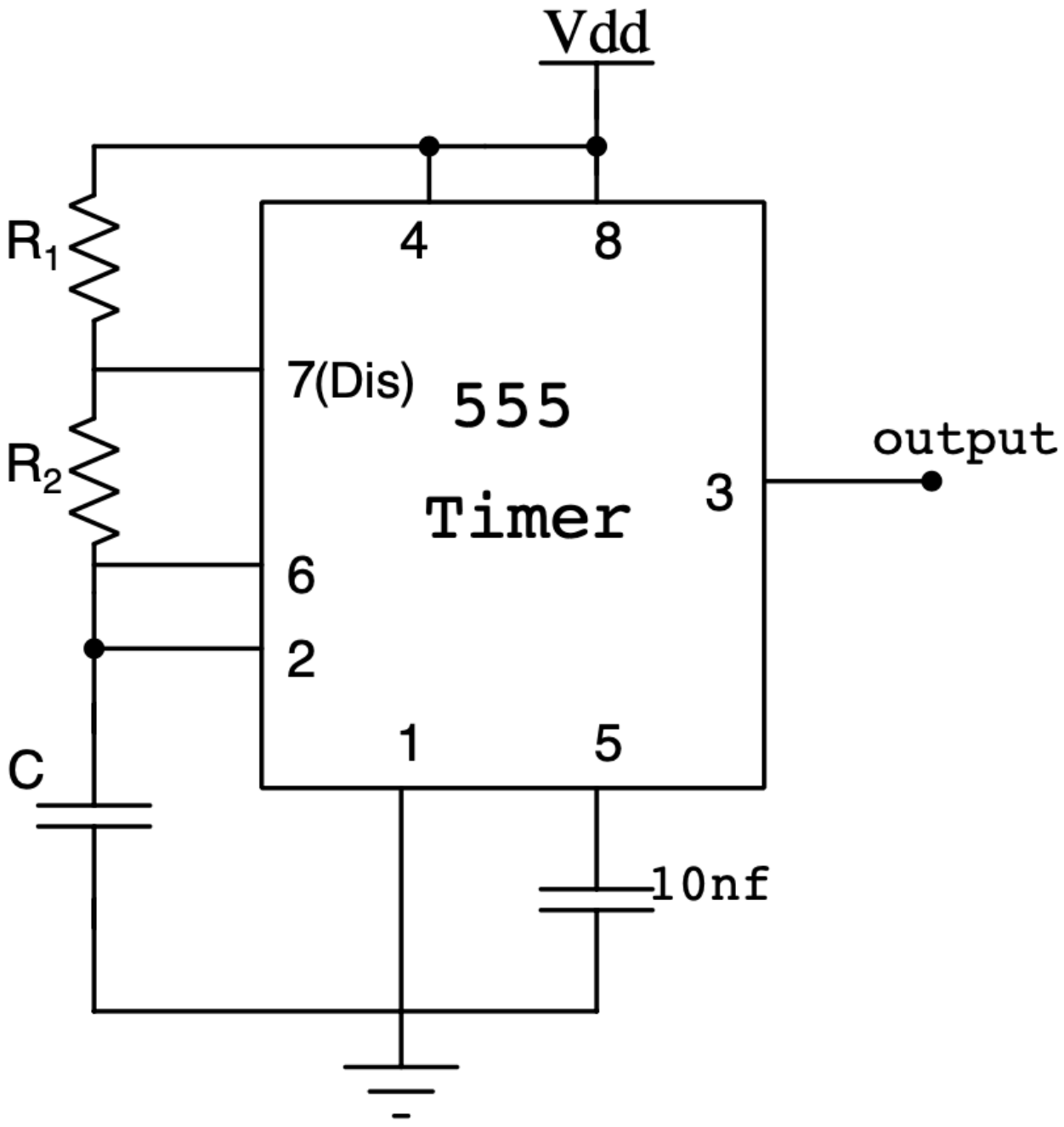
On-Time T_{on} and Off-Time T_{off} are

$$T_{on} = \ln(2)(R_1 + R_2)C \qquad T_{off} = \ln(2)R_2C$$

and the Duty cycle is given by

$$Duty\ cycle = \frac{R_2}{R_1 + 2R_2} * 100\%$$

As an example, for achieving 1kHz oscillations with 50% Duty-Cycle, Select $R_1 = 1k\Omega$, $R_2 = 72k\Omega$ and $C = 10nF$. Or select any other combination of R_1, R_2 and C. But ensure R_1, R_2 is **not** less than $1k\Omega$.


Vdd
R1
R2
C
4
8
7(Dis)
555
Timer
6
2
1
5
3
output
10nf